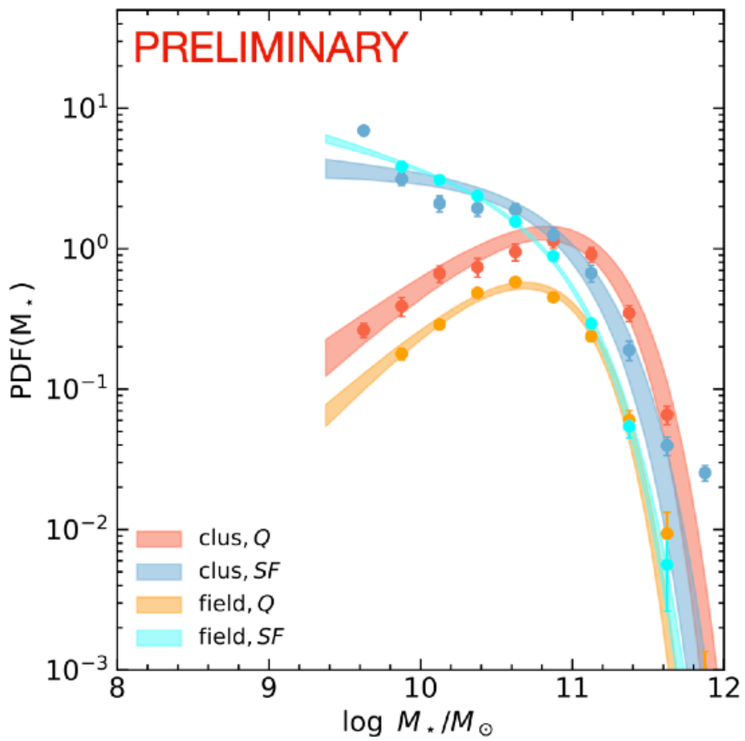
Galaxy stellar-mass functions

New method : use SED fitting $\mathrm{PDF}(z, M_\star)$ and $p_Q(z, M_\star)$ in the inference of cluster galaxy stellar-mass functions

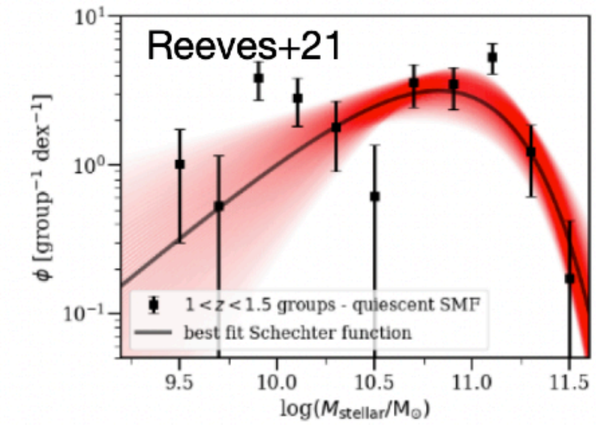
→ important for Euclid/LSST samples were uncertainty will be dominated by systematics



Application to REFINE

- Competitive constraints for the group environment at $z > 1$

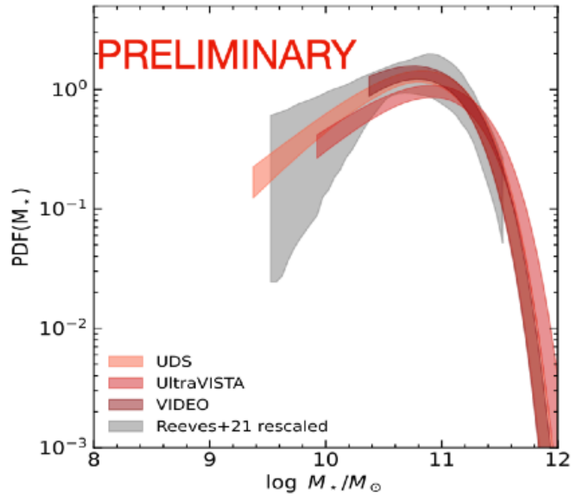
$1 < z < 1.5$



GOGREEN

21 groups

X-ray selected



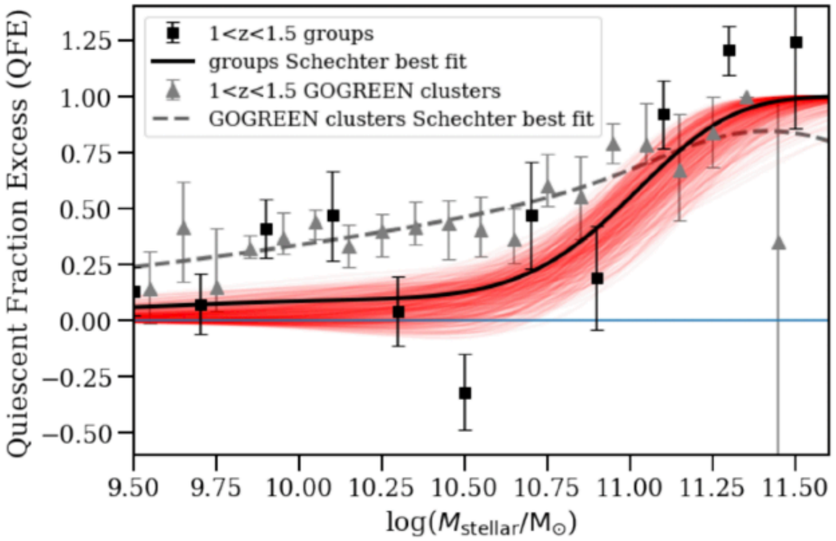
REFINE

328 candidate groups

$\mathrm{purity}(z) > 50\,\%$

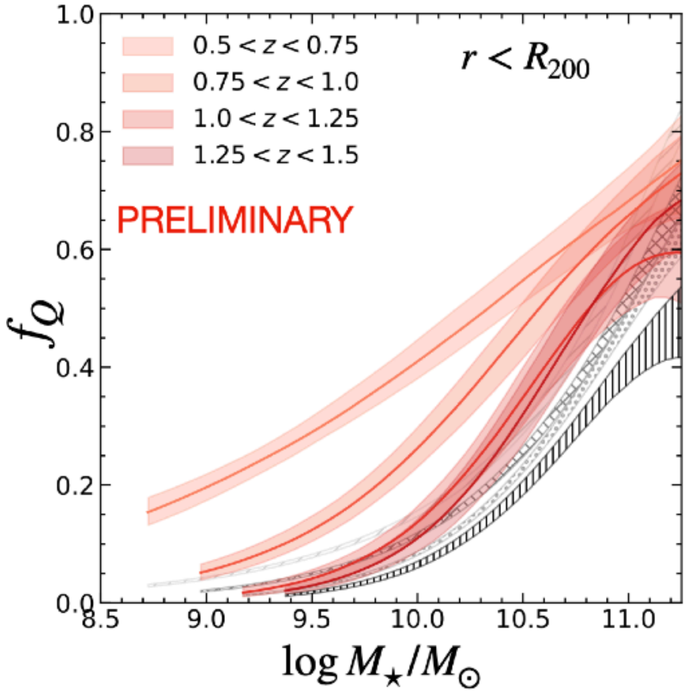
QFE as a function of stellar-mass and redshift

Reeves+21

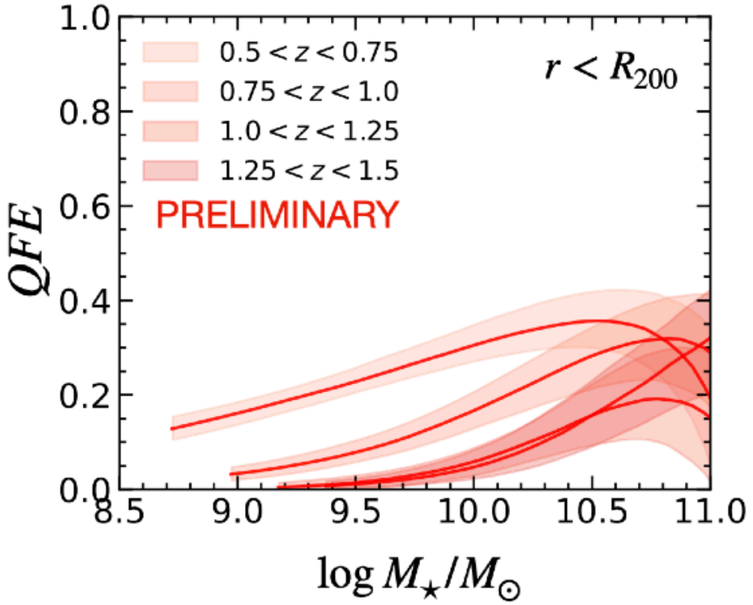


Groups not efficient at quenching lower mass galaxies at $1 < z < 1.5$

REFINE



- The quenched fraction at $M_\star < 10.5$ increases with decreasing redshift $z <$ in galaxy groups



- The QFE $M_\star$ dependence in groups steepens at $z > 1$ due to a lack of quenched low mass galaxies at lower masses — similar to ZFOURGE

In progress — Hierarchical model to fit the redshift, group mass and radial distance on the full REFINE sample